Synthetic BOAMP Benchmark Technical Report

Latent-truth generator, validation framework, and current readiness

Current independent study

End-to-end refresh: 2026-07-27

Abstract

The synthetic BOAMP benchmark creates a controlled procurement-renewal world with hidden truth relations, corrupts the observable fields to resemble BOAMP, and evaluates linkage algorithms against sealed truth. The current benchmark version is `v0_3_temporal_candidate_revision`. The validation framework reports `PASS_WITH_WARNINGS`; critical internal, specification-recovery, candidate-environment, privacy, and algorithm-utility gates pass, but conditionals, temporal runway, hidden-truth difficulty, and robustness carry warnings. Two noncritical SIRET missing/present metrics fail. The five-level readiness assessment is authoritative: preliminary synthetic modeling is allowed with limitations, while controlled algorithm comparison, final algorithm ranking, and validated benchmark release currently fail.

1 Current Decision

Table 1: Five-level readiness assessment from the current run.

Decision	Status
`PIPELINE_TECHNICALLY_VALID`	`PASS_WITH_LIMITATIONS`
`READY_FOR_PRELIMINARY_MODELING`	`PASS_WITH_LIMITATIONS`
`READY_FOR_CONTROLLED_ALGORITHM_COMPARISON`	`FAIL`
`READY_FOR_FINAL_ALGORITHM_RANKING`	`FAIL`
`READY_FOR_VALIDATED_SYNTHETIC_BENCHMARK_RELEASE`	`FAIL`

The benchmark may be used for preliminary synthetic-only linker debugging and pipeline integration. It must not be cited as ready for controlled algorithm comparison, final ranking, or validated release.

2 Selected v0.3 Candidate Parameters

The selected central-provisional candidate setting is saved in `candidate_revision_selected_parameters.csv`, `generation_metadata.json`, and `config/synthetic/scenarios/central_provisional.yaml`. The key selected values are:

- recurrence propensity multiplier: 0.88;
- near-window share: 1.00;
- hard-negative alignment rate: 0.34;
- scoped buyer affinity share: 0.102;
- high scoped-need probability: 0.74.

3 Candidate Environment

Table 2: Layer-1 algorithm source-scope candidate environment.

Scope	Sources	Zero-cand.	Mean	Median	p75	p90/p95
Real Layer 1	3,380	40.7%	3.21	1	4	9.1 / 14
v0.2 synthetic	354	90.4%	0.15	0	0	0 / 1
v0.3 synthetic	422	38.6%	3.45	1	6	11 / 13

The current v0.3 candidate environment passes the predefined gates: source-count ratio 1.196, zero-candidate absolute difference 2.05 percentage points, p75 absolute difference 2, p90 absolute difference 1.9, p95 absolute difference 1, cap-reached rate 0, and buyer-key zero-candidate rates within tolerance.

4 Validation Framework

Table 3: Current gate summary.

Gate	Critical	Status
Internal integrity	yes	PASS
Specification recovery	yes	PASS
Marginals	no	WARNING
Conditionals	yes	WARNING
Temporal	yes	WARNING
Candidate environment	yes	PASS
Missingness/text/identifier	no	WARNING, 2 metric failures
Buyer activity	no	WARNING
Missingness structure	no	PASS
Names and identifiers	no	WARNING
Text	no	WARNING
Privacy	yes	PASS
Hidden-truth difficulty	yes	WARNING
Algorithm utility	yes	PASS
Robustness	no	WARNING

The two metric-level failures are noncritical SIRET missing/present rate gaps: the synthetic SIRET present rate is 31.6% versus 27.2% real, an absolute difference of 4.37 percentage points against a 2 percentage point tolerance. The validation manifest therefore remains `PASS_WITH_WARNINGS`, not `FAIL`, but readiness still blocks controlled comparison because the failures and robustness warnings are not acceptable for that use.

5 Hidden Truth and Algorithm Utility

The benchmark keeps latent truth separate from observed notices. The observed layer passes checks for explicit truth columns, hidden-truth alias column names, exact or normalized truth-ID values, and direct MD5/SHA1/SHA256 hashes of hidden truth identifiers.

Table 4: Current hidden-truth difficulty and probe utility.

Quantity	Value
In-scope true successor pairs	120
Production-window blocking recall	0.467
36-month blocking recall	0.942
Production-window pairs quality	0.038
Best current probe pair-F1	0.352
Deterministic top-1 pair-F1	0.308
Score-threshold all-ranks pair-F1	0.352
Fellegi–Sunter pair-F1	0.192

6 Robustness and Replay

The current artifact set includes 10 generated seeds each for `central_provisional`, `easier`, `moderate`, `difficult`, and `stress`; `clean_sanity` remains a single-seed smoke test. Canonical replay passes for the central artifact and all-artifact replay passes for 51 generated artifacts.

Robustness remains warning-level. Central production blocking completeness has mean 0.399, standard deviation 0.128, and one non-pass seed. Central probe-headroom pair-F1 has mean 0.281, standard deviation 0.110, and one non-pass seed. Probe rankings are unstable: the top probe is rank 1 in 50% of central/moderate seeds and 58% of all benchmark-scenario artifacts.

7 Source State and Reproducibility

The source-state manifest records Git HEAD `40dfad87881560ebd8b6dae91fa6397d545bc9cd` and hashes the current dirty source/artifact overlay. `dirty_state_overlay.tar.gz` packages that overlay, and `dirty_state_overlay_verification.json` verifies that it applies to a clean Git HEAD snapshot with no missing files or hash mismatches. This is auditability, not release approval.

The current machine-readable evidence is:

- `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/generation_manifest.jso`
- `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/candidate_environment_s`
- `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/validation_framework/va`
- `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/readiness/readiness_dec`
- `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/validation_framework/re`

8 Next Step

The valid next step is to improve the remaining observable-fidelity and robustness gaps before using the benchmark for controlled algorithm comparison. Any algorithm work performed before that point should be described as preliminary synthetic-only debugging or pipeline integration.